# LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

## Ch 8 Test (Set B): Heredity

*Chapter / Topic: Heredity*

Name: _____ Roll No.: _____ Date: _____

**Time: 50 minutes** **Max. Marks: 25**

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**General instructions:**

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Show crosses clearly with gametes and offspring wherever asked.

### SECTION A

6 marks

*Choose the correct option.*

**Q1.** Plants with round seeds (RR) are crossed with plants with wrinkled seeds (rr). The $F_1$ seeds are: **[1]**

(a) all wrinkled (b) all round
(c) half round and half wrinkled (d) 3 round : 1 wrinkled

**Q2.** How many pairs of chromosomes are there in human body cells? **[1]**

(a) 21 (b) 22
(c) 23 (d) 46

**Q3.** A child has the sex chromosomes XX. The child is: **[1]**

(a) a boy (b) impossible to predict
(c) either a boy or a girl (d) a girl

**Q4.** In the $F_2$ generation of Mendel's dihybrid cross, what fraction of plants show both recessive traits (wrinkled and green)? **[1]**

(a) 1/16 (b) 3/16
(c) 1/4 (d) 9/16

**Q5.** Mendel carried out his experiments on: **[1]**

(a) fruit flies (b) garden peas
(c) wheat (d) mice

**Q6.** **Assertion (A):** There is very little variation among offspring produced by asexual reproduction. **Reason (R):** In asexual reproduction, small errors in DNA copying are the only source of variation. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

### SECTION B

4 marks

**Q7.** Why were all the $F_1$ plants tall, with no medium-height plants, when Mendel crossed pure tall and pure short pea plants? **[2]**

**OR**

Define genotype and phenotype, with one example each.

**Q8.** A man with blood group A marries a woman with blood group O, and their daughter has blood group O. Is this enough to tell which trait, A or O, is dominant? Explain. **[2]**

## SECTION C

6 marks

**Q9.** A pea plant with violet flowers (Vv) is crossed with a plant with white flowers (vv). **[3]**

**(i)** Write the gametes produced by each parent. **[1]**

**(ii)** Write the genotypes of the offspring and their ratio. **[1]**

**(iii)** Write the phenotypic ratio. **[1]**

**Q10.** How do Mendel's experiments show that (i) traits may be dominant or recessive, and (ii) traits are inherited independently? **[3]**

**OR**

How is the number of chromosomes kept the same from one generation to the next? What is the role of germ cells?

## SECTION D

5 marks

**Q11.** Answer the following: **[5]**

**(i)** Explain Mendel's dihybrid cross between round yellow (RRYY) and wrinkled green (rryy) plants, up to the $F_2$ phenotypes and their ratio. **[3]**

**(ii)** Which new combinations appear in $F_2$, and what do they prove? **[2]**

**OR**

(a) Explain how sex is determined in (i) human beings, (ii) some reptiles, (iii) snails. [3]
(b) Is the sex of a human child decided by chance? Explain. [2]

## SECTION E

4 marks

*Read the passage and answer the questions that follow.*

**Q12.** Earlobes can be free or attached. Assume that free earlobes (F) are dominant over attached earlobes (f). Both of Arjun's parents have free earlobes, but Arjun has attached earlobes. **[4]**

**(i)** What is Arjun's genotype? **[1]**

**(ii)** What must the genotypes of his parents be? **[1]**

**(iii)** What is the probability that their next child will have attached earlobes? Show the cross. **[2]**

*~ All the best! ~*